



Combination of Alpha-Lipoic Acid and Auraptene Induces Apoptosis and Prevents Proliferation of the Human U-87 Glioblastoma Cells

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Abstract

Glioblastoma multiforme is a fatal brain tumor with a poor survival of 14.6 months. Radiation therapy in combination with temozolomide is the widely used treatment for this aggressive cancer. In recent decades, herbal medicines such as alpha-lipoic acid and auraptene have attracted worldwide attention as combination therapy alternatives due to their potential antitumor effects in various types of cancers and animal models. However, there is insufficient information about the anticancer effects of this combination on glioblastoma multiforme. Therefore, the cytotoxic effects of the α -lipoic acid/auraptene combination on glioblastoma cell proliferation and apoptosis were investigated. The anti-proliferative effects of the α -lipoic acid/auraptene combination on U87 cells was examined via MTT colorimetric assay. Bax and Bcl-2 transcript expression levels were assessed by real-time quantitative PCR. The intracellular level of reactive oxygen species was examined. Finally, the cell cycle distribution and apoptosis of U87 cells after treatment with α -lipoic acid/auraptene were investigated via flow cytometry. Our results showed that this combination suppressed U87 cell viability dose-dependently. The combinations tested induced apoptosis, reflected in decreased Bcl-2 and increased Bax expression at the mRNA level. Therapy with this combination also enhanced the generation of ROS, effectively causing G2/M cell cycle arrest in U87 cells.

Keywords Combination therapy · Antiproliferative effect · 7-Geranyloxy coumarin · Organosulfur compound · Reactive oxygen species

Introduction

Glioblastoma multiforme (GBM) is an invasive and malignant brain neoplasm originating from the astrocytes of the central nervous system. The disease affects more men and those with a family history of unclear inherited risk factors. The average age at diagnosis is 64 years and the median survival time following diagnosis is approximately 14.6 months (Liu et al. 2023). Major molecular traits that distinguish GBM include extensive cancer cell invasion, metastasis, and proliferation (Xie et al. 2014). The currently recommended treatment strategy includes the complete resection of lesions followed by radiotherapy and chemotherapy with temozolomide (Pouratian and Schiff 2010). Due to tumors' poor resection, recurrence, and resistance to radiotherapy and chemotherapy, alternative and adjuvant treatments, such as therapy with natural and herbal substances, have recently come under investigation (Tagde et al. 2021). Additionally, due to their low toxicity and extensive therapeutic benefits, herbal medications

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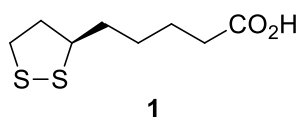
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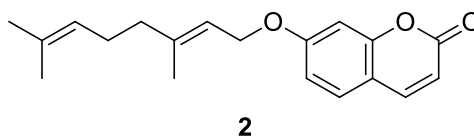
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have been researched globally (Yuan et al. 2016; Ye et al. 2018). Thus, in this study, we investigated the effects of the α -lipoic acid/auraptene combination on cell growth and apoptosis in U87 cells with GBM.

α -Lipoic acid (ALA, **1**) and its reduced form dihydrolipoic acid are herb-derived organosulfur compounds that exhibit promising effects in cancer treatment (Gorąca et al. 2011). The utilization of α -lipoic acid in dietary anti-aging supplements and multivitamin formulations has increased due to its antioxidant and anti-diabetic properties (Tripathi et al. 2023). Considering its favorable effects and low toxicity, ALA can be regarded as a desirable candidate in cancer treatment (Costantino et al. 2014; Cremer et al. 2006). α -Lipoic acid possesses a wide range of therapeutic effects, including antioxidant, anti-diabetic, anti-obesity, hypotensive, and anti-inflammatory. It is also involved in the regeneration of glutathione and vitamin C and E metabolism (Gomes and Negrato 2014; Akbari et al. 2018; Najafi et al. 2022). Studies have shown that ALA can serve as an inhibitor of proliferation and angiogenesis and can induce apoptosis (Yoo et al. 2013; Attia et al. 2020). Previous reports have demonstrated the inhibitory effect of ALA on cell proliferation, exerted through targeted activation of the AMPK signaling cascade and subsequent attenuation of AKT signaling (Jeon et al. 2016). The effect of ALA on apoptosis is closely associated with increased expression of pro-apoptotic proteins such as Bax and decreased expression of the anti-apoptotic protein Bcl-2 (Na et al. 2009). Recently, ALA has been reported to inhibit metastasis and invasiveness by reducing MMP-2/9 expression and activity in cancer cells. This is because these enzymes are involved in proteolytic collagen degradation of extracellular matrix components (Halbersztadt et al. 2006; Conlon and Murray 2019).



Auraptene (AUR, **2**), 7-geranyloxycoumarin, is a monoterpene coumarin ether isolated from members of the Rutaceae and Apiaceae families (Afshari et al. 2019; Curini et al. 2006). Several pharmacological properties have been described for compound **2**, such as anti-diabetic, anti-genotoxic, anti-inflammatory, antioxidant, and immunomodulatory effects (Genovese and Epifano 2011). Auraptene can serve as an alternative therapy due to its favorable properties and low toxicity in the treatment of cancers (Mousavi et al. 2015; Vakili et al. 2017). Based on recent research, AUR exerts inhibitory effects on proliferation, tumorigenesis, and growth by affecting transcription factors, growth factors, and cytokines. In addition, AUR can induce apoptosis and gene-modulating cellular proliferation.



Glioblastoma multiforme is the most common cancer that arises in the brain and affects 3 out of 100,000 people each year. Given that current treatments are not effective enough and can have serious side effects, newer therapies with greater efficacy and therapeutic potential for this common and aggressive cancer should be sought. Therefore, we investigated the effects of a combination of α -lipoic acid and auraptene on U87 cancer cells' proliferation and apoptosis.

Materials and Methods

Materials

The human malignant glioblastoma cell line (U87) was procured from Pasteur Institute (National Cell Bank of Iran), Tehran, Iran. Auraptene (**2**, purity of 96%; Mashhad, Iran; Lot number: T49500203) and α -lipoic acid (**1**, purity of 98%, Lot number: BCBR6063V) were acquired from Sigma-Aldrich (St Louis, MO, USA). Trypan blue, propidium iodide, penicillin/streptomycin, dimethyl sulfoxide (DMSO), and trypsin-EDTA were also purchased from Sigma-Aldrich (St. Louis, Mo, USA). The [3-(4,5 dimethyl thiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT) was obtained from Thermo Fisher (Thermo Fisher Scientific Inc., USA). High-glucose Dulbecco's modified eagle's medium (DMEM) and fetal bovine serum (FBS) were procured from Gibco (Grand Island, NY, USA). The dichloro-dihydro-fluorescein diacetate (DCFDA)/H2DCFDA-cellular reactive oxygen species (ROS) detection assay kit was supplied from Abcam (Cambridge, UK). The annexin V-FITC/propidium iodide apoptosis assay kit was procured from BioLegend (San Diego, CA, USA).

MTT Assay

The U87 cells were grown in high-glucose DMEM comprising 10% FBS and 1% penicillin-streptomycin antibiotic and were retained in a humidified incubator at 37 °C with 5% CO₂. The medium was daily changed. The cytotoxicity of ALA and AUR was investigated by the colorimetric MTT assay. In brief, 1×10^4 U87 cells/well were seeded in a 96-well plate and incubated overnight. Then, the cells were exposed to different concentrations of AUR (0–2.68 mM) and ALA (0–100 mM) for 24 h. After removing the medium, the cells were incubated with 100 μ l FBS-containing fresh medium with 0.5 mg/ml MTT included; after 3 h

of incubation, the medium was aspirated and 200 μ l DMSO was added to each well to dissolve the formazan crystals. Eventually, the optical density at 570 nm was measured using a Stat FAX303 plate reader. The IC_{50} values were calculated by nonlinear regression analysis in GraphPad Prism 9. In this method, the cells were treated with temozolomide (reference drug) as the positive control. All the samples were tested in triplicate.

Combination Index Determination

The cells were exposed to AUR/ALA with concentrations of 1/2 and 1/4 IC_{50} , IC_{50} , and 2 and 4 times of the compounds' IC_{50} , alone or combined, for 24 h (Table S1). After treatment, the MTT tests were performed as described in the previous section. The combination index (CI) was calculated using GraphPad Prism and Calcsyn 2.0. $CI < 1$ represents a synergism effect, $CI = 1$ demonstrates an additive effect, and $CI > 1$ shows an antagonism effect (Chou 2011).

Quantitative Real-Time PCR

To examine the expression of Bax and Bcl-2 after ALA/AUR treatment in U87 cells, total cellular RNA was extracted with a total RNA extraction kit (Pars Tous, Iran). RNA purity and concentration were then determined using a Nanodrop spectrophotometer. Subsequently, complementary DNA (cDNA) was prepared using an Easy cDNA Synthesis Kit (Pars Tous Co., Iran) according to the manufacturer's instructions. The qRT-PCR reactions were run on a Roche real-time thermal cycler using SYBR Green PCR master mix (SMOBIO, Taiwan). The primers for Bax (Bcl-2) are listed in Table S2. All the data were normalized using GAPDH as a control gene, and fold changes were calculated using the $2^{-\Delta\Delta CT}$ method (Livak and Schmittgen 2001).

Reactive Oxygen Species Assay

Intracellular ROS aggregation was measured using a commercially available ROS detection kit (DCFDA/H2DCFDA). U87 cells (25×10^3 /well) were cultured in 96-well plates and incubated overnight. The cells were then washed with 100 μ l of 1X buffer and incubated with 100 μ l of the H2DCFDA solution (25 μ M) for 60 min at 37 °C in the dark. After being washed again, the cells were incubated with AUR (0 and 0.2 mM), ALA (0 and 12.5 mM), and a combination of the two (0.2 mM AUR and 12.5 mM ALA) for 1, 4, 24, and 48 h (triplicates for each sample). *N*-acetylcysteine (NAC) (10 mM) was used as the positive control. Finally, fluorescence was assessed (excitation/emission: 485/535 nm) using a FACScan fluorescence plate reader (Becton Dickinson, San Joes, USA).

Cell Cycle Analysis

To examine the effect of the ALA/AUR combination on cell cycle distribution, we used flow cytometry in U87 cells. For this purpose, 7×10^5 U87 cells/well were seeded in a 6-well plate and incubated overnight. The cells were then treated with AUR (0 and 0.2 mM), ALA (0 and 12.5 mM), and ALA/AUR combinations (0.2 mM AUR and 12.5 mM ALA) for 24 h. After the treatment, the cells were harvested using trypsin, resuspended in cold PBS, and fixed with 70% ethanol at 4 °C for 2 h. In the next step, the cells were incubated in PBS containing 100 μ g/ml ribonuclease A (RNase A) and 0.1% triton X-100 for 20 min at 37 °C. Finally, the samples were stained with 400 μ l of propidium iodide solution (1 mg/ml) and incubated for 15 min in the dark. The data were then analyzed by flow cytometry.

Apoptosis Analysis

The effect of the ALA/AUR combination on apoptosis was examined using the FITC Annexin V Apoptosis Detection kit (BioLegend, CA, USA) with propidium iodide. U87 cells (7×10^5) cultured in 6-well plates and incubated at 37 °C for 24 h. The cells were then treated with AUR (0 and 0.2 mM), ALA (0 and 12.5 mM), and their combination (0.2 mM AUR and 12.5 mM ALA) for 24 h. Subsequently, the cells were trypsinized and resuspended in 100 μ l annexin-V binding buffer. Then, the cells were stained with 5 μ l of FITC annexin V and 10 μ l of propidium iodide solution and incubated for 15 min at room temperature in the dark. Finally, 400 μ l of the annexin V binding buffer was added to each sample, and the data were analyzed using flow cytometry.

Statistical Analysis

The data were analyzed using Calcsyn 2.0 and GraphPad Prism ® 9.0. One-way analysis of variance (ANOVA) followed by Dunnett's test were used to compare the values and quantified ratios obtained from different groups. All the experiments were performed in triplicate and the results are expressed as mean $\pm$ standard deviation (SD), with $p < 0.05$ considered statistically significant.

Results and Discussion

U87 Cell Viability

U87 cells were exposed to various concentrations of AUR and ALA for 24 h, and the viability of the treated cells was compared with that of untreated controls. Our results clearly indicated the ability of ALA and AUR to decrease U87 cell

proliferation dose-dependently (Fig. 1). The half-maximal inhibitory concentration (IC_{50}) values of AUR and ALA in U87 cells were 0.4 and 25 mM, respectively.

Proliferation of U87 Cells

U87 cells were exposed to various combined concentrations of ALA and AUR, as shown in Table S1. Among the different combination doses, AUR/ALA 0.2/12.5 mM had a CI < 1 and approximate viability of 50%, so this combination ratio was chosen for further experiments (Fig. 2B). As shown in Fig. 2A, AUR/ALA significantly decreased cell viability compared to the AUR ($p < 0.0001$, Fig. 2A).

Inhibition of Apoptosis-related Genes Expression

The effects of ALA, AUR, and their combination on the expression of genes involved in apoptosis (Bax and Bcl-2) were evaluated using qRT-PCR. After treatment, a significant decrease in Bcl-2 mRNA levels and an increase in Bax mRNA levels were observed in U87 cells compared to control (Fig. 3). Furthermore, ALA/AUR combination therapy exerted greater inhibitory effects on Bax and Bcl-2 expression compared to ALA or AUR monotherapy ($p < 0.0001$).

ROS Levels

To investigate the role of ROS in cytotoxicity induced by ALA, AUR, and their combination, we measured

the generation of intracellular ROS levels in U87 cells. ALA, AUR, and the ALA-AUR combination significantly increased intracellular ROS levels in U87 cells compared with the control group (Fig. 4), which possibly caused oxidative damage in U87 cells.

G2/M Cell Cycle

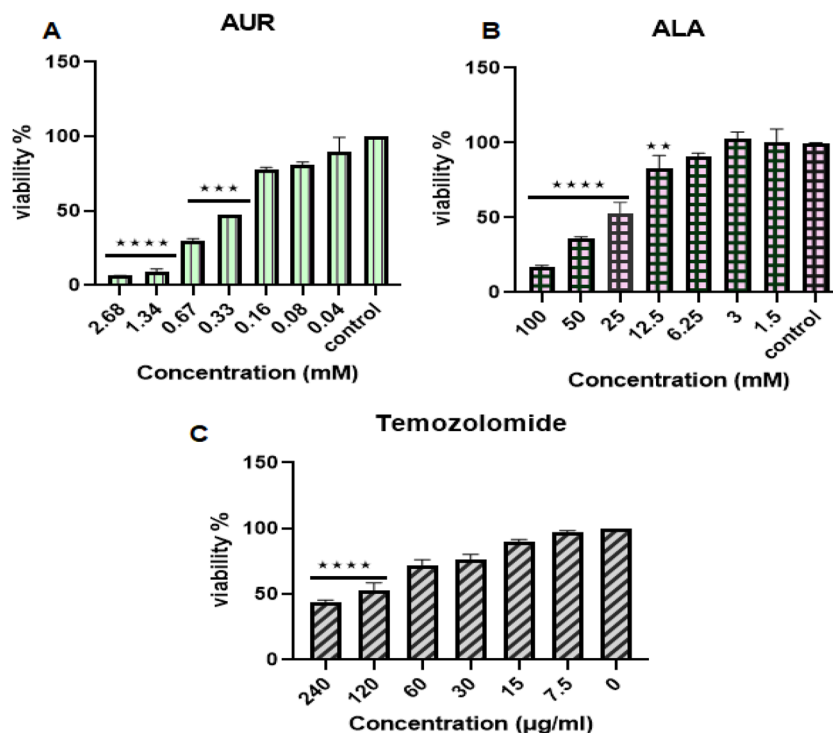
Based on Fig. 5, treatment of cells with the ALA/AUR combination significantly increased the sub-G1 area compared to that of the control group and led to a subsequent cell cycle arrest in the G2/M phase.

Apoptosis

As the second step of our study, annexin V-FITC/propidium iodide staining was applied to further validate the effect of the ALA/AUR combination on cell apoptosis. As shown in Fig. 6, treatment with the ALA/AUR combination significantly increased the percentage of apoptotic cells compared to the control group.

We investigated the effect of the ALA/AUR combination on cell proliferation and apoptosis in U87 cells. Notably, the ALA/AUR combination suppressed the proliferation and induced apoptosis in U87 cells. Analysis of the relative expression of apoptosis-involved genes (Bax and Bcl-2) revealed that the ALA/AUR combination enhanced apoptosis in U87 cells through increased intracellular ROS and cell cycle disruption. Therefore, the ALA/AUR combination can

Fig. 1 Cell viability assay of U87 cells treated with auraptene (A), α -lipoic acid (B), and temozolomide (C). Cell toxicity was evaluated by the MTT assay. Both compounds decreased the viability of U87 cells dose-dependently, with IC_{50} values of 0.4 mM for auraptene and 25 mM for α -lipoic acid ($p < 0.05$). The data are represented as the mean $\pm$ SD of three independent experiments. ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$ show significant differences compared with the control group. Abbreviations: ALA, α -lipoic acid; AUR, auraptene



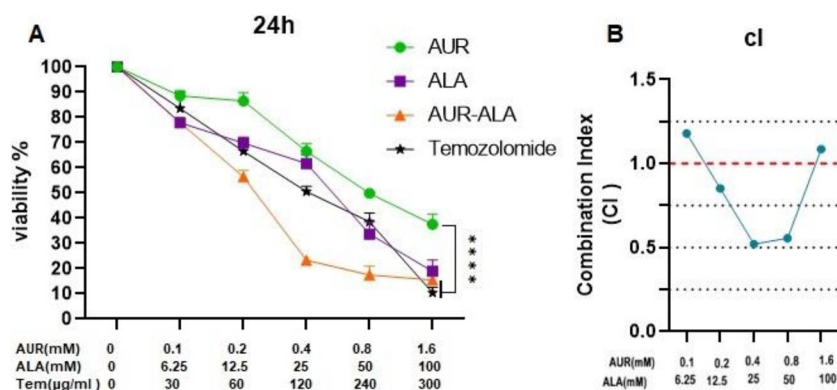


Fig. 2 Cell viability assay and determination of combination index (CI) for various auraptene (2)/α-lipoic acid (1) combination concentrations. **(A)** MTT results indicated a dose-dependent reduction in the viability of U87 cells after α-lipoic acid/auraptene combination treatment. **(B)** CI plot for the auraptene/α-lipoic acid combination treatment. The optimum synergistic effect was observed in the com-

bination comprising 0.04 mM auraptene and 25 mM α-lipoic acid. CI < 1 indicates the synergistic effect of two compounds in the corresponding concentrations. The data are represented as the mean ± SD of three independent experiments (*****p* < 0.0001). Abbreviations: ALA, α-lipoic acid; AUR, auraptene; Tem, Temozolomide

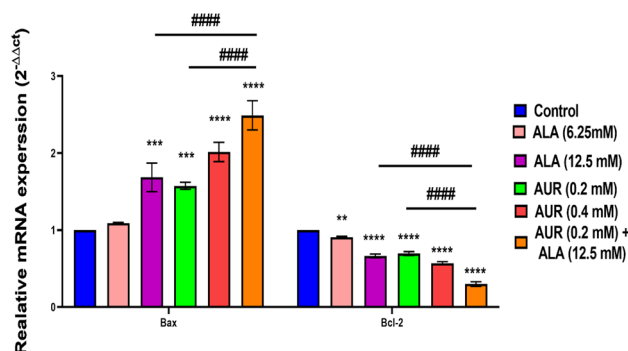


Fig. 3 Assessment of Bax and Bcl-2 mRNA expression with the qRT-PCR method. The cells were treated with AUR 0.2 mM, α-lipoic acid 12.5 mM, and auraptene (0.2 mM)/α-lipoic acid (12.5 mM) combination for 24 h. The results showed a significant decrease in mRNA of Bcl-2 expression and a significant increase in mRNA of Bax expression. The data are presented as mean ± SD; ***p* < 0.01, ****p* < 0.001, and *****p* < 0.0001 as compared to the control group, and #####*p* < 0.0001 as compared to the α-lipoic acid/auraptene combination group. Abbreviations: ALA, α-lipoic acid; AUR, auraptene

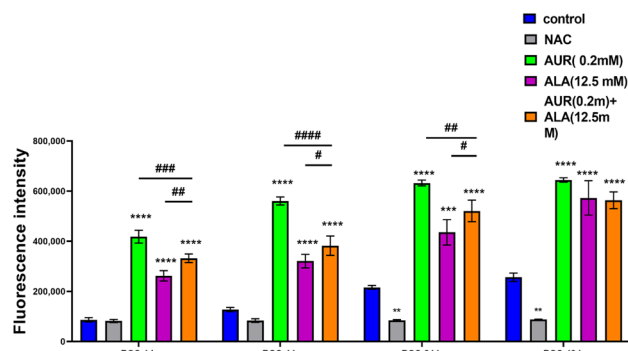


Fig. 4 Determination of the effects of α-lipoic acid (1), auraptene (2), and auraptene/α-lipoic acid combination on intracellular ROS in U87 cells. The cells were treated with auraptene (0.2 mM), α-lipoic acid (12.5 mM), and their combination (0.2 mM auraptene-12.5 mM α-lipoic acid). The data are presented as mean ± SD; ***p* < 0.01, ****p* < 0.001, and *****p* < 0.0001 as compared to the control group. ##*p* < 0.01, ###*p* < 0.001, and #####*p* < 0.0001 as compared to the α-lipoic acid/AUR combination group. Abbreviations: ALA, α-lipoic acid; AUR, auraptene

be a suitable candidate for restoring dysregulated cell growth and death patterns in tumor cells.

Several reports have described the anti-proliferative properties of ALA or AUR in different cancer cells and models (Dozio et al. 2010; McConnell et al. 2018; Yang et al. 2019; Ebrahimi et al. 2023b). Consistent with previous findings, Kuban-Jankowska et al. (2017) found that ALA potently attenuated MCF-7 breast cancer cell viability and cell development by inhibiting the phosphatase activity of PTP1B and SHP2. Peng et al. (2020) found that ALA suppressed lung cancer cell viability through mTOR-dependent autophagy inhibition. Michikoshi et al. (2013) also reported

that ALA effectively prevented the proliferation of non-small cell lung cancer cells, which was attributed to its inhibitory effect on the activation of growth factor receptors such as Met and EGFR (Michikoshi et al. 2013). A combination of ALA treatment and paclitaxel, a widely used cocktail chemotherapy, can suppress breast cancer cell growth and proliferation by modulating NF-κB signaling (Li et al. 2015). Similarly, several reports have highlighted the anti-proliferative properties of AUR. Moussavi et al. (2018) demonstrated that AUR combined with hyperthermia significantly decreased cell survival and stimulated the expression of the P21 gene in HT-29 colon cancer cells as compared to control

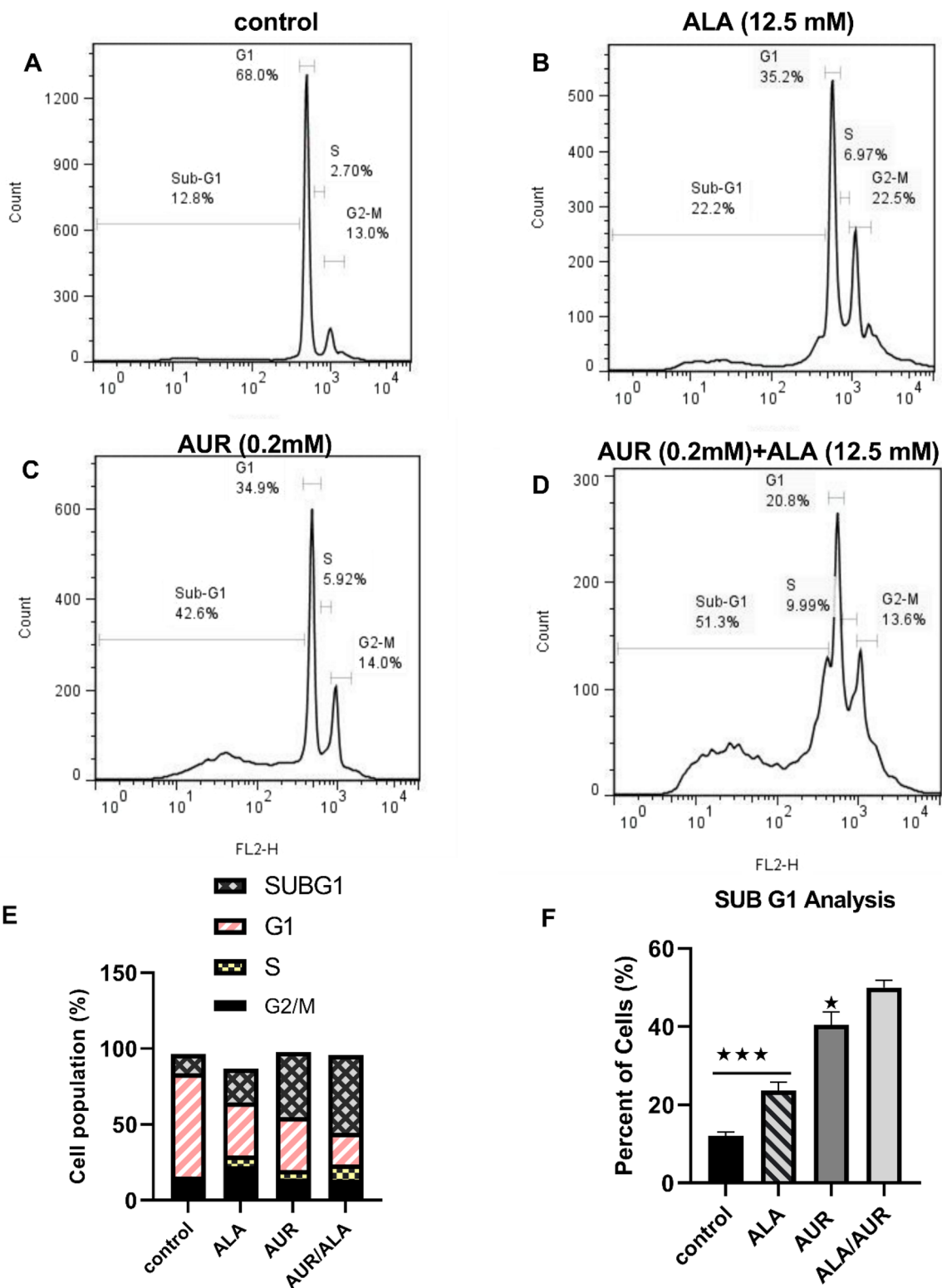


Fig. 5 Cell cycle distribution of U87 cells following α -lipoic acid (1), auraptene (2), and α -lipoic acid/auraptene treatment. Representative histograms of cell cycle distribution depict apoptosis in U87 cells treated with (B) α -lipoic acid, (C) auraptene, (D) α -lipoic acid/auraptene combination compared with the (A) control group ($p < 0.05$). (E) Analysis of cell population in each cell cycle

phase using FlowJo software. (F) Percentage of cells in the sub-G1 population relative to the total number of cells. The data are presented as mean $\pm$ SD; * $p < 0.05$ and *** $p < 0.001$ show significant differences compared to the control group; Abbreviations: ALA, α -lipoic acid; AUR, auraptene

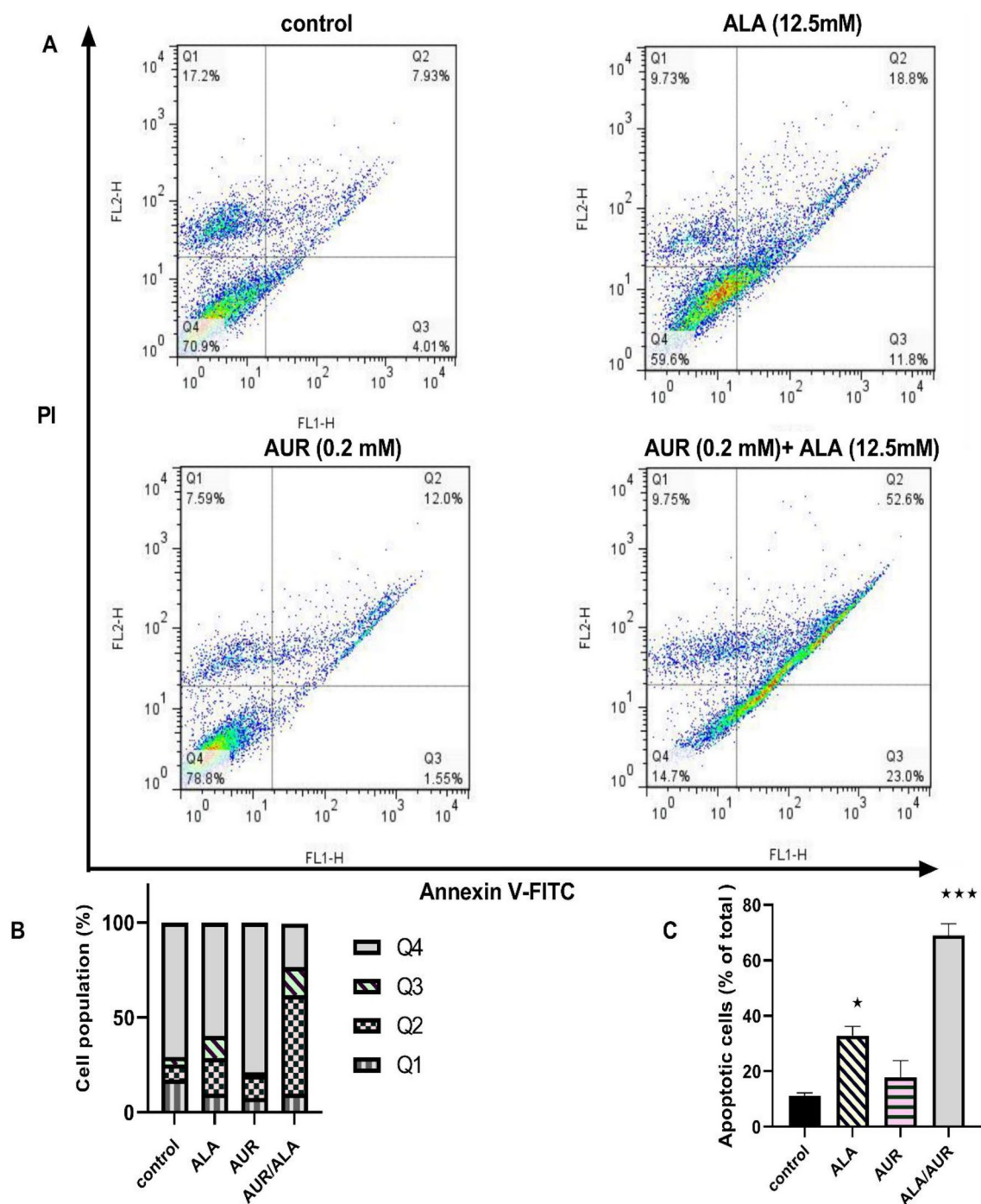


Fig. 6 Flow cytometry analysis of α -lipoic acid, auraptene, and the α -lipoic acid/auraptene-induced apoptosis in U-87 cells stained with annexin V-FITC/propidium iodide. **(A)** The cells were treated with α -lipoic acid, auraptene, and the α -lipoic acid/auraptene combination and stained with annexin V-FITC/propidium iodide. **(B)** Analysis of cell population in each cell cycle phase relative to total phases, **(C)**

A statistical graph of annexin V-FITC/propidium iodide staining is depicted. The data averages for each time point were calculated using the results of three independent experiments. The data are presented as (mean $\pm$ SD). * $p < 0.05$ and *** $p < 0.001$ show significant differences compared to the control group

cells. According to Sakata et al. (2004), dietary intake of AUR in rats effectively suppressed *N,N*-diethylnitrosamine-induced hepatocarcinogenesis by inhibiting cell proliferation and stimulating apoptosis. Consistent with the antiproliferative properties of AUR, Lee et al. (2017) reported that AUR restricted the proliferation of PC3 and DU145 prostate cancer cells time-dependently. Another study by AUR also reported cell proliferation inhibition and suppression of cyclin D1 expression (Krishnan et al. 2009). Consistent with these studies, our results showed that ALA, AUR, and the ALA-AUR combination significantly suppressed the proliferation of GBM U87 cells, compared with untreated control cells, with the best effects obtained via the combination treatment.

Recent studies have shown that carcinogenesis is closely related to the failure to regulate multiple biological pathways and the subsequent failure of apoptosis which results in cell cycle dysregulation (Goldar et al. 2015). The ALA/AUR combination resulted in cell cycle arrest in the G2/M phase and decreased cell proliferation. Consistently, Choi et al. (2021) observed that ALA alone and/or in combination with radiotherapy stimulated cell cycle arrest in the G2/M phase of MDA-MB-231 breast cancer cells. This drug also down-regulated the expression of cell cycle regulators such as the P21 protein. Another study found that ALA caused a significant dose-dependent aggregation in the G1 phase of the cell cycle dose-dependently, accompanied by a decrease in the S and G2 phases. Furthermore, ALA increased the expression of P27kip1 which functions as a cell cycle regulator (Dozio et al. 2010). In a study conducted by Moon et al. (2015) using AUR, activation of P53 and suppression of mTOR signaling in SUN-1 cells caused cell cycle arrest and stimulated apoptosis. Finally, Ebrahimi et al. (2023a) showed that AUR exerts an inhibitory effect on colon cancer growth and metastasis by inducing cell cycle arrest and apoptosis.

We evaluated the effect of the ALA/AUR combination on the externalization of phosphatidylserine during apoptosis by annexin V-FITC staining. Furthermore, the induction of apoptosis was facilitated by modulating the expression of apoptosis-associated genes such as Bax and Bcl-2 (Burlacu 2003). In this regard, a report by Na et al. (2009) on ALA showed increased Bax mRNA and protein expression, decreased Bcl-2 mRNA and protein expression, and the subsequent induction of caspase-3 in breast cancer. Lee et al. (2017) demonstrated that apoptosis induction by AUR is associated with the activation of caspase-3/9, upregulation of Bax translational expression levels, and decreased regulation of Bcl-2 and Mcl-1 levels. Consistent with these results, our study demonstrated that the ALA/AUR combination increased the transcriptional expression of Bax and decreased the transcriptional expression of Bcl-2 in U87 GBM cells, indicating that apoptosis is associated with these genes.

Another report found that increased ROS production induced apoptosis in cancer cells (Perillo et al. 2020). ROS generation occurs through regular oxygen metabolism. However, cytotoxicity happens in tumor cells when ROS levels exceed a certain level (Hsin et al. 2008; Uguz et al. 2012). According to Kim et al. (2012), ALA induces sub-G1 phase cell apoptosis by increasing ROS levels in lung cancer cells. Afshari et al. (2019) found that treatment with AUR increased ROS levels 8 and 24 h after treatment, resulting in decreased cell viability of glioma cells (Afshari et al. 2019). Based on our results, ALA, AUR, and the ALA/AUR combination increased the intracellular ROS levels and possibly caused oxidative damage in U87 cells.

Conclusion

Our results suggested that the combination of ALA/AUR has a significant anti-proliferative effect in U87 cells and can induce the apoptosis associated with increased expression of apoptosis-related genes and intracellular ROS levels in U87 cells. Therefore, it may provide effective chemotherapy for glioblastoma. Further studies should elucidate the anti-cancer mechanisms of the ALA-AUR combination.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s43450-023-00440-4>.

Author contributions MS and AI contributed to the study's conception and design. AI carried out all the experiments. AI, AS, MJN, FM, and AA contributed to the interpretation of the results. The manuscript was drafted by AI, and revisions were made by AI and MS for publication. The corresponding author declares that all the listed authors meet the authorship criteria.

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Data Availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics Approval All experiments were performed in accordance with the guidelines of the Ethics Committee of Kerman University of Medical Sciences (IR.KMU.AH.REC.1400.053).

Conflict of Interest The authors declare no conflict of interest.

References

- Afshari AR, Jalili-Nik M, Soukhtanloo M, Ghorbani A, Sadeghnia HR, Mollazadeh H, Karimi Roshan M, Rahmani F, Sabri H, Vahedi MM, Mousavi SH (2019) Auraptene-induced cytotoxicity mechanisms in human malignant glioblastoma (U87) cells:

- role of reactive oxygen species (ROS). *Excli J* 18:576–590. <https://doi.org/10.17179/excli2019-1136>
- Akbari M, Ostadmohammadi V, Tabrizi R, Mobini M, Lankarani KB, Moosazadeh M, Heydari ST, Chamani M, Kolahdooz F, Asemi Z (2018) The effects of alpha-lipoic acid supplementation on inflammatory markers among patients with metabolic syndrome and related disorders: a systematic review and meta-analysis of randomized controlled trials. *Nutr Metab* 15:39. <https://doi.org/10.1186/s12986-018-0274-y>
- Attia M, Essa EA, Zaki RM, Elkordy AA (2020) An overview of the antioxidant effects of ascorbic acid and alpha lipoic acid (in liposomal forms) as adjuvant in cancer treatment. *Antioxidants* 9:359. <https://doi.org/10.3390/antiox9050359>
- Burlacu A (2003) Regulation of apoptosis by Bcl-2 family proteins. *J Cell Mol Med* 7:249–257. <https://doi.org/10.1111/j.1582-4934.2003.tb00225.x>
- Choi HS, Kim JH, Jang SJ, Yun JW, Kang KM, Jeong H, Ha IB, Jeong BK (2021) Synergistic tumoricidal effects of alpha-lipoic acid and radiotherapy on human breast cancer cells via HMGB1. *Cancer Res Treat* 53:685–694. <https://doi.org/10.4143/crt.2020.1015>
- Chou TC (2011) The mass-action law based algorithm for cost-effective approach for cancer drug discovery and development. *Am J Cancer Res* 1:925–954
- Conlon GA, Murray GI (2019) Recent advances in understanding the roles of matrix metalloproteinases in tumour invasion and metastasis. *J Pathol* 247:629–640. <https://doi.org/10.1002/path.5225>
- Costantino M, Guaraldi C, Costantino D, De Grazia S, Unfer V (2014) Peripheral neuropathy in obstetrics: efficacy and safety of α -lipoic acid supplementation. *Eur Rev Med Pharmacol Sci* 18:2766–2771. <https://doi.org/10.13140/RG.2.1.1948.8485>
- Cremer DR, Rabeler R, Roberts A, Lynch B (2006) Safety evaluation of alpha-lipoic acid (ALA). *Regul Toxicol Pharmacol* 46:29–41. <https://doi.org/10.1016/j.yrtph.2006.06.004>
- Curini M, Cravotto G, Epifano F, Giannone G (2006) Chemistry and biological activity of natural and synthetic prenyloxycoumarins. *Curr Med Chem* 13:199–222. <https://doi.org/10.2174/092986706775197890>
- Dozio E, Ruscica M, Passafaro L, Dogliotti G, Steffani L, Marthyn P, Pagani A, Demartini G, Esposti D, Frascini F, Magni P (2010) The natural antioxidant alpha-lipoic acid induces p27(Kip1)-dependent cell cycle arrest and apoptosis in MCF-7 human breast cancer cells. *Eur J Pharmacol* 641:29–34. <https://doi.org/10.1016/j.ejphar.2010.05.009>
- Ebrahimi S, Mostafavi-Pour Z, Khazaei M, Nazari SE, Jamshidi ST, Soukhtanloo M (2023a) Suppression of metastasis by *Citrus auraptene* in a mouse model of colorectal cancer. *Rev Bras Farmacogn* 33:182–190. <https://doi.org/10.1007/s43450-022-00351-a>
- Ebrahimi S, Soukhtanloo M, Mostafavi-Pour Z (2023b) Anti-tumor effects of auraptene through induction of apoptosis and oxidative stress in a mouse model of colorectal cancer. *Tissue Cell* 81:102004. <https://doi.org/10.1016/j.tice.2022.102004-b>
- Genovese S, Epifano F (2011) Auraptene: a natural biologically active compound with multiple targets. *Curr Drug Targets* 12:381–386. <https://doi.org/10.2174/138945011794815248>
- Goldar S, Khaniani MS, Derakhshan SM, Baradaran B (2015) Molecular mechanisms of apoptosis and roles in cancer development and treatment. *Asian Pac J Cancer Prev* 16:2129–2144. <https://doi.org/10.7314/apjcp.2015.16.6.2129>
- Gomes MB, Negrato CA (2014) Alpha-lipoic acid as a pleiotropic compound with potential therapeutic use in diabetes and other chronic diseases. *Diabetol Metab Syndr* 6:80. <https://doi.org/10.1186/1758-5996-6-80>
- Goraça A, Huk-Kolega H, Piechota A, Kleniewska P, Ciejka E, Skibska B (2011) Lipoic acid - biological activity and therapeutic potential. *Pharmacol Rep* 63:849–858. [https://doi.org/10.1016/s1734-1140\(11\)70600-4](https://doi.org/10.1016/s1734-1140(11)70600-4)
- Halbersztadt A, Haloń A, Pajak J, Robaczyński J, Rabczynski J, St Gabryś M (2006) The role of matrix metalloproteinases in tumor invasion and metastasis. *Ginek Pol* 77:63–71
- Hsin YH, Chen CF, Huang S, Shih TS, Lai PS, Chueh PJ (2008) The apoptotic effect of nanosilver is mediated by a ROS- and JNK-dependent mechanism involving the mitochondrial pathway in NIH3T3 cells. *Toxicol Lett* 179:130–139. <https://doi.org/10.1016/j.toxlet.2008.04.015>
- Jeon MJ, Kim WG, Lim S, Choi HJ, Sim S, Kim TY, Shong YK, Kim WB (2016) Alpha lipoic acid inhibits proliferation and epithelial mesenchymal transition of thyroid cancer cells. *Mol Cell Endocrinol* 419:113–123. <https://doi.org/10.1016/j.mce.2015.10.005>
- Kim JI, Cho SR, Lee CM, Park ES, Kim KN, Kim HC, Lee HY (2012) Induction of ER stress-mediated apoptosis by α -lipoic acid in A549 cell lines. *Korean J Thorac Cardiovasc Surg* 45:1–10. <https://doi.org/10.5090/kjtes.2012.45.1.1>
- Krishnan P, Yan KJ, Windler D, Tubbs J, Grand R, Li BD, Aldaz CM, McLarty J, Kleiner-Hancock HE (2009) *Citrus auraptene* suppresses cyclin D1 and significantly delays *N*-methyl nitrosourea induced mammary carcinogenesis in female Sprague-Dawley rats. *BMC Cancer* 9:25. <https://doi.org/10.1186/1471-2407-9-259>
- Kuban-Jankowska A, Gorska-Ponikowska M, Wozniak M (2017) Lipoic acid decreases the viability of breast cancer cells and activity of PTP1B and SHP2. *Anticancer Res* 37:2893–2898. <https://doi.org/10.21873/anticancer.11642>
- Lee JC, Shin EA, Kim B, Kim BI, Chitsazian-Yazdi M, Iranshahi M, Kim SH (2017) Auraptene induces apoptosis via myeloid cell leukemia 1-mediated activation of caspases in PC3 and DU145 prostate cancer cells. *Phytother Res* 31:891–898. <https://doi.org/10.1002/ptr.5810>
- Li BJ, Hao XY, Ren GH, Gong Y (2015) Effect of lipoic acid combined with paclitaxel on breast cancer cells. *Genet Mol Res* 14:17934–17940. <https://doi.org/10.4238/2015.December.22.18>
- Liu D, Dai X, Ye L, Wang H, Qian H, Cheng H, Wang X (2023) Nanotechnology meets glioblastoma multiforme: emerging therapeutic strategies. *Wiley Interdiscip Rev Nanomed Nanobiotechnol* 15:e1838. <https://doi.org/10.1002/wnan.1838>
- Livak KJ, Schmittgen TD (2001) Analysis of relative gene expression data using real-time quantitative PCR and the $2^{-\Delta\Delta C_T}$ Method. *Methods* 25:402–408. <https://doi.org/10.1006/meth.2001.1262>
- McConnell DD, McGreevy JW, Williams MN, Litofsky NS (2018) Do anti-oxidants vitamin D(3), melatonin, and alpha-lipoic acid have synergistic effects with temozolomide on cultured glioblastoma cells? *Medicines* 5:58. <https://doi.org/10.3390/medicines5020058>
- Michikoshi H, Nakamura T, Sakai K, Suzuki Y, Adachi E, Matsugo S, Matsumoto K (2013) α -Lipoic acid-induced inhibition of proliferation and met phosphorylation in human non-small cell lung cancer cells. *Cancer Lett* 335:472–478. <https://doi.org/10.1016/j.canlet.2013.03.008>
- Moon JY, Kim H, Cho SK (2015) Auraptene, a major compound of supercritical fluid extract of phalsak (*Citrus Hassaku* Hort ex Tanaka), induces apoptosis through the suppression of mTOR pathways in human gastric cancer SNU-1 cells. *Evid-Based Compl Alt Med* 2015:402385. <https://doi.org/10.1155/2015/402385>
- Mousavi SH, Davari AS, Iranshahi M, Sabouri-Rad S, Tayarani Najaran Z (2015) Comparative analysis of the cytotoxic effect of 7-prenyloxycoumarin compounds and herniarin on MCF-7 cell line. *Avicenna J Phytomed* 5:520–530. <https://doi.org/10.22038/ajp.2015.4355>
- Moussavi M, Haddad F, Matin MM, Iranshahi M, Rassouli FB (2018) Efficacy of hyperthermia in human colon adenocarcinoma cells is improved by auraptene. *Biochem Cell Biol* 96:32–37. <https://doi.org/10.1139/bcb-2017-0146>

- Na MH, Seo EY, Kim WK (2009) Effects of alpha-lipoic acid on cell proliferation and apoptosis in MDA-MB-231 human breast cells. *Nutr Res Pract* 3:265–271. <https://doi.org/10.4162/nrp.2009.3.4.265>
- Najafi N, Mehri S, Ghasemzadeh Rahbardo M, Hosseinzadeh H (2022) Effects of alpha lipoic acid on metabolic syndrome: a comprehensive review. *Phytother Res* 36:2300–2323. <https://doi.org/10.1002/ptr.7406>
- Peng P, Zhang X, Qi T, Cheng H, Kong Q, Liu L, Cao X, Ding Z (2020) Alpha-lipoic acid inhibits lung cancer growth via mTOR-mediated autophagy inhibition. *FEBS Open Bio* 10:607–618. <https://doi.org/10.1002/2211-5463.12820>
- Perillo B, Di Donato M, Pezone A, Di Zazzo E, Giovannelli P, Galasso G, Castoria G, Migliaccio A (2020) ROS in cancer therapy: the bright side of the moon. *Exp Mol Med* 52:192–203. <https://doi.org/10.1038/s12276-020-0384-2>
- Pouratian N, Schiff D (2010) Management of low-grade glioma. *Curr Neurol Neurosci Rep* 10:224–231. <https://doi.org/10.1007/s11910-010-0105-7>
- Sakata K, Hara A, Hirose Y, Yamada Y, Kuno T, Katayama M, Yoshida K, Zheng Q, Murakami A, Ohigashi H, Ikemoto K, Koshimizu K, Tanaka T, Mori H (2004) Dietary supplementation of the citrus antioxidant auraptene inhibits N, N-diethylnitrosamine-induced rat hepatocarcinogenesis. *Oncology* 66:244–252. <https://doi.org/10.1159/000078001>
- Tagde P, Tagde P, Tagde S, Bhattacharya T, Garg V, Akter R, Rahman MH, Najda A, Albadrani GM, Sayed AA, Akhtar MF, Saleem A, Altyar AE, Kaushik D, Abdel-Daim MM (2021) Natural bioactive molecules: An alternative approach to the treatment and control of glioblastoma multiforme. *Biomed Pharmacother* 141:111928. <https://doi.org/10.1016/j.biopha.2021.111928>
- Tripathi AK, Ray AK, Mishra SK, Bishen SM, Mishra H, Khurana A (2023) Molecular and therapeutic insights of alpha-lipoic acid as a potential molecule for disease prevention. *Rev Bras Farmacogn* 33:272–287. <https://doi.org/10.1007/s43450-023-00370-1>
- Uguz AC, Cig B, Espino J, Bejarano I, Naziroglu M, Rodríguez AB, Pariente JA (2012) Melatonin potentiates chemotherapy-induced cytotoxicity and apoptosis in rat pancreatic tumor cells. *J Pineal Res* 53:91–98. <https://doi.org/10.1111/j.1600-079X.2012.00974.x>
- Vakili T, Iranshahi M, Arab H, Riahi B, Roshan NM, Karimi G (2017) Safety evaluation of auraptene in rats in acute and subacute toxicity studies. *Regul Toxicol Pharmacol* 91:159–164. <https://doi.org/10.1016/j.yrtph.2017.10.025>
- Xie Q, Mittal S, Berens ME (2014) Targeting adaptive glioblastoma: an overview of proliferation and invasion. *Neuro Oncol* 16:1575–1584. <https://doi.org/10.1093/neuonc/nou147>
- Yang Y, Fang E, Luo J, Wu H, Jiang Y, Liu Y, Tong S, Wang Z, Zhou R, Tong Q (2019) The antioxidant alpha-lipoic acid inhibits proliferation and invasion of human gastric cancer cells via suppression of STAT3-mediated MUC4 gene expression. *Oxid Med Cell Longev* 2019:3643715. <https://doi.org/10.1155/2019/3643715>
- Ye J, Zhang R, Wu F, Zhai L, Wang K, Xiao M, Xie T, Sui X (2018) Non-apoptotic cell death in malignant tumor cells and natural compounds. *Cancer Lett* 420:210–227. <https://doi.org/10.1016/j.canlet.2018.01.061>
- Yoo TH, Lee JH, Chun HS, Chi SG (2013) α -Lipoic acid prevents p53 degradation in colon cancer cells by blocking NF- κ B induction of RPS6KA4. *Anticancer Drugs* 24:555–565. <https://doi.org/10.1097/CAD.0b013e32836181eb>
- Yuan H, Ma Q, Ye L, Piao G (2016) The traditional medicine and modern medicine from natural products. *Molecules* 21:559. <https://doi.org/10.3390/molecules21050559>

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